

Kinetic computational model of explosive emission center initiation by interaction of dense plasma with microprotrusion

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A kinetic one-dimensional spherically symmetric model of heating of microprotrusion (with curvature radius of 1 micrometer) on the cathode under dense plasma was developed. The model of PIC/DSMC type takes into account the main types of elastic and inelastic collisions of particles in the plasma as well as evaporation and thermo-field electron emission from the cathode. The surrounding plasma parameters in the model are fixed on the boundary placed at the certain distance from the microprotrusion. The applied voltage drop is also fixed on the same boundary. The further evolutions of the plasma, current, and cathode temperature are calculated selfconsistently taking into account the influence of the plasma generated during the process. The calculations have shown that under the certain condition the thermal runaway is developed, and cathode surface temperature reaches the critical one.

1. Introduction

The mechanism of the explosive centre ignition on cathode surface is one of the key points of the vacuum arc theory [1]. There is a strong opinion that in the vacuum condition the explosive centre ignition is provided by the explosion of microprotrusion, which is heated by the field electron emission current up to the temperature above the critical one [1]. However, when the cathode is covered and screened by plasma the electric field at the cathode surface is not strong enough to provide necessary emission current density. In this case the particle fluxes from the plasma to cathode play the significant role. These fluxes must be strong enough to heat and explode the microprotrusion.

The microprotrusion heating by the plasma action was quite intensively studied previously (see Ref. [2]-[4] for example and for further references). The possibility of microprotrusion heating up to critical temperature in the necessary time range (several nanoseconds) was shown. The main heating processes in that case were the ion bombardment and the Joule heating by the thermo-field emission current. The influence of the plasma electron flux was insignificant. Thus, the previous works could, in principle, explain the explosive center ignition under the plasma. However, the calculations made previously were not selfconsistent. The plasma parameters and cathode voltage drop were the task parameters, which were fixed during the microprotrusion heating. One of the approaches which allow taking the plasma and the plasma sheath evolution into consideration is the kinetic PIC/MC method. This method allows one to model

the microprotrusion heating and the correspondent plasma production and evolution selfconsistently. However, the method is highly demanding computer power. Due to this, the present model was restricted by 1D spherical symmetry geometry.

2. Physical model

The model describes the metal (copper) microprotrusion (Fig.1) which is heated by both plasma metal interaction and joule heating. The model describes the evaporation from the metal surface and correspondent secondary plasma generation. The secondary plasma generation and evolution is described by the kinetic model of 1D3V PIC/DSMC type. The kinetic model takes into account the main types of inelastic and elastic collisions (including the coulomb ones) of particles in the plasma as well as evaporation and thermo-field electron emission from the cathode. Similar kinetic model was used before in Ref. [5].

The microprotrusion shape is roughly depicted in Fig. 1. The microprotrusion consists of a hemisphere with 1 μm radius and a neck. The total height of the microprotrusion is 2 μm . The neck merges the metal substrate. The area of the neck

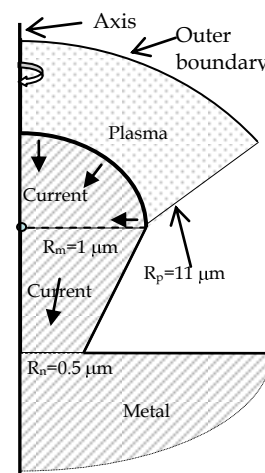


Fig. 1. Sketch of task geometry

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14. ABSTRACT AA kinetic one-dimensional spherically symmetric model of heating of microprotrusion (with curvature radius of 1 micrometer) on the cathode under dense plasma was developed. The model of PIC/DSMC type takes into account the main types of elastic and inelastic collisions of particles in the plasma as well as evaporation and thermo-field electron emission from the cathode. The surrounding plasma parameters in the model are fixed on the boundary placed at the certain distance from the microprotrusion. The applied voltage drop is also fixed on the same boundary. The further evolutions of the plasma, current, and cathode temperature are calculated selfconsistently taking into account the influence of the plasma generated during the process. The calculations have shown that under the certain condition the thermal runaway is developed, and cathode surface temperature reaches the critical one.					
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footpoint is eight times less than the area of the top hemisphere. Thus, the current density at the neck footpoint will be eight times as high as the current density at the surface of the top hemisphere. This artificial shape is chosen to roughly model (due to 1D geometry restriction) the microprotrusion of droplet-on-neck type. The microprotrusions of that type appear at the crater rim of burning cathode spot. It is the most popular site for the ignition of the next explosive emission center [1].

The plasma domain looks geometrically like a spherical cone. The spherical cone is made by cone generator with half-angle at vertex $\pi/4$, an internal radius equal to 1 μm , and an outer radius equal to 11 μm . Thus, the plasma surface interaction is calculated only at the top hemisphere of the microprotrusion. The lateral surface of the microprotrusion neck and surface of the metal substrate don't interact with the plasma due to 1D geometrical restriction. The outer border of the plasma domain is constituted by an imaginary surface within the external plasma which can be penetrated by plasma particles from both sides. A ghost cell with plasma parameters identical to those obtained for the plasma in front of the outer boundary is used. Hence, some of the electrons and ions flow into the area of calculation through the outer plasma boundary.

The initial conditions of the task are the following: the microprotrusion has temperature of 2800 K; the metal substrate has temperature of 400 K; plasma is homogeneous with fixed ion density in the range of $10^{17} - 10^{18} \text{ cm}^{-3}$, ion charge state $z=2$, ion temperature of 2 eV, electron temperature of 5 eV. The plasma parameters are taken from the simulation results of plasma expansion at the functioning of the cathode spot emission center [6]. Electrical potential at the top hemisphere of the microprotrusion equals zero. The potential at the outer boundary of plasma domain U_0 is equals to the certain value which is kept constant during the calculations. After starting the calculations the

evolutions of the plasma, current, and cathode temperature are calculated selfconsistently taking into account the influence of the plasma generated during the process. The change of the microprotrusion shape is not taken into account. The surface heating, joule heating, and thermal conduction are taken in consideration for the calculation of the microprotrusion temperature evolution.

3. Results

Initial conditions (n_i , U_0) used in the simulations as well as some results, like time to ignition and reached surface temperature, are summarized in Table I.

The behavior of the microprotrusion top surface temperature in course of time is shown in Fig.2. It is seen that the curves have singularity. Thus, the thermal instability takes place. The surface temperature goes to infinity according to hyperbolic law when time approaches the certain moment. In reality, the calculation is stopped when the temperature reaches the critical one (8390 K for Cu). After the reaching this temperature the microprotrusion should start to explode. The modeling of the explosion demands the use of additional technique; it is out of task here. Thus, it is assumed that the new explosive emission center is ignited when the temperature reaches the critical temperature. The calculation showed that time to the ignition (or delay times) for the used task parameters (Table I) varies from 1.51 ns (condition 1) to 5.07 ns (condition 5). For conditions 6, 7 the surface temperature does not reach the critical temperature. The temperature reaches saturation value, which is slightly above 5000 K. Thus, in framework of the model there is no cathode spot ignition for the conditions 6, and 7. It is seen that

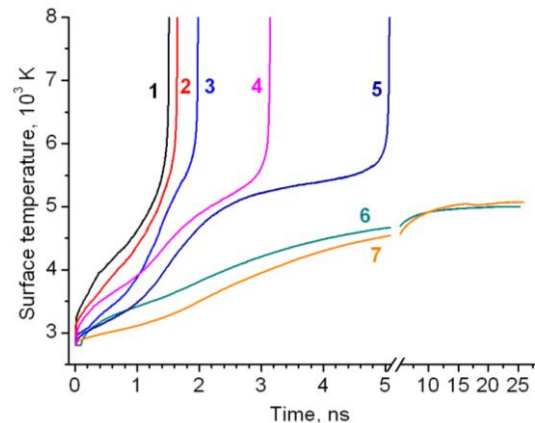


Fig. 2. Temperature at the top surface of microprotrusion for different conditions. Conditions for n_i and U_0 for cases 1-7 are summarized in Table I. Other initial parameters: $T_e=5 \text{ eV}$, $T_i=2 \text{ eV}$, $z=2$ in all cases.

TABLE I

INITIAL CONDITIONS OF CALCULATION AND RESULTS

#	Ion density, n_i , cm^{-3}	Potential, U_0 , V	Time to ignition, ns	Max surface temperature, K
1	10^{18}	18	1.51	8390*
2	10^{18}	20	1.65	8390*
3	10^{18}	25	1.98	8390*
4	$5 \cdot 10^{17}$	20	3.14	8390*
5	$5 \cdot 10^{17}$	25	5.07	8390*
6	10^{17}	20	-	5002
7	10^{17}	25	-	5072

* - calculations are stopped at this temperature.

the delay time increases with the voltage U_0 increasing.

Temperature distribution inside the microprotrusion for different time instants is shown in Fig. 3. It is seen that the temperature maximum is located at the top surface all the time. The graph of current density components (Fig. 4) also demonstrates the singularity character. At the ignition moment the total current density reaches value of $\sim 6 \cdot 10^7$ A/cm². It is quite low value in comparison with the thermal runaway provoked by joule heating [1]-[3]. It is seen that the ion current density is comparatively small, but electron emission current density and backward plasma electron current density reach value of about 10^9 A/cm². Close to ignition time, these two current densities compensate each other with about 90% accuracy, so that the total current density is about one order of magnitude less than the aforementioned current components.

Temporal evolution of the heat flux components near the surface is shown in Fig. 5. The curves also have the singular character. It is clear that the most important heat sources are the electron emission cooling and the heating by the backward plasma electrons. However, the heating by the plasma electron is stronger. The joule heating plays insignificant role in the used configuration. The input of energy flux of neutrals (including the evaporation cooling) to the total energy flux is negligible almost all the time, because the evaporation is suppressed by the plasma pressure. The evaporation energy input becomes essential only when the surface temperature approach to the critical one.

Behaviour of plasma density with time is shown in Fig. 6. The density near the microprotrusion surface increases with time. At the ignition time the ion density reaches the value of $\sim 5 \cdot 10^{20}$ cm⁻³. The density grows because of two reasons: evaporation from the metal surface with further ionization and plasma flow from the outer boundary of the spherical cone to inside due to near-cathode plasma sheath formation. As it can be seen from Fig. 7, most of the time the plasma flows to the microprotrusion. The plasma reaches quite high velocity of $\sim 7 \cdot 10^5$ cm/s. That is approximately two times higher than the sound velocity at corresponding spatial position. According to the electric potential distribution (Fig. 8) the ion flow is accelerated from the outer boundary to the microprotrusion reaching the supersonic velocity. These concurrent plasma streams produce the

plasma compression around the microprotrusion accompanied by the plasma density growth. This mechanism of the plasma density rise near the microprotrusion predominates over the evaporation mechanism during most of the calculation time. Close to the surface the plasma is decelerated due to

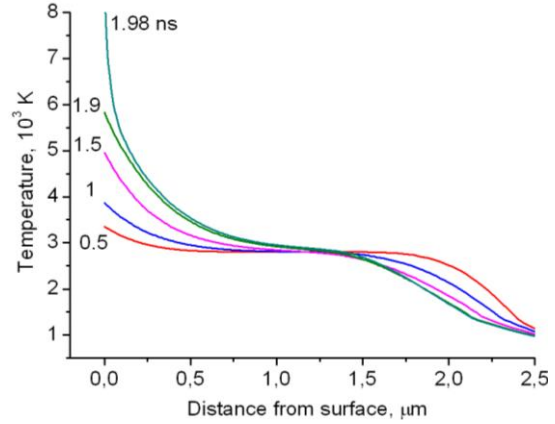


Fig. 3. Temperature distribution in the microprotrusion (along Axis in Fig. 1) for different time instants (# 3. Table I). The zero distance is at the top surface.

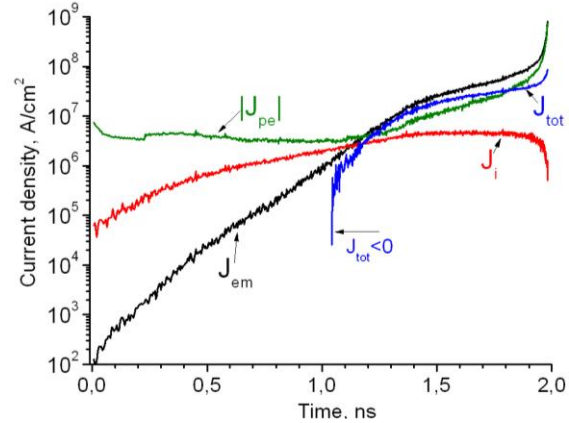


Fig. 4. Current density components at top surface of microprotrusion (# 3. Table I). J_{em} -electron emission current; J_i - ion current, J_{pe} - backward electron current from plasma to surface; $J_{tot}=J_{em}+J_i-|J_{pe}|$ - total current.

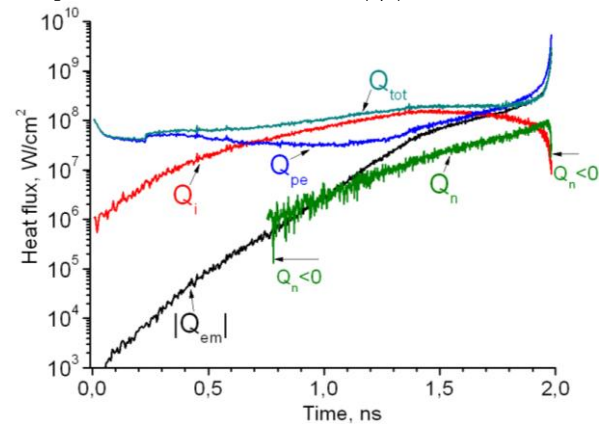


Fig. 5. Heat flux components at the microprotrusion surface (# 3, Table I); Q_{pe} - electron heat flux from plasma to microprotrusion; Q_i -ion heat flux; Q_{em} - electron emission heat flux; Q_n - heat flux of neutrals (including evaporation); $Q_{tot}=Q_{pe}+Q_i+Q_n-|Q_{em}|$ - total heat flux

interaction with the metal vapor and secondary plasma (the plasma produced from evaporated metal). When the surface temperature reaches approximately 6000 K, the average plasma velocity near the surface becomes positive. The secondary plasma starts expanding to the gap. This is a beginning of cathode plasma jet formation.

The distribution of electric potential in the plasma gap is shown in Fig. 8. It is seen that the cathode drop is developed but the value of the drop is well below the floating potential for the given

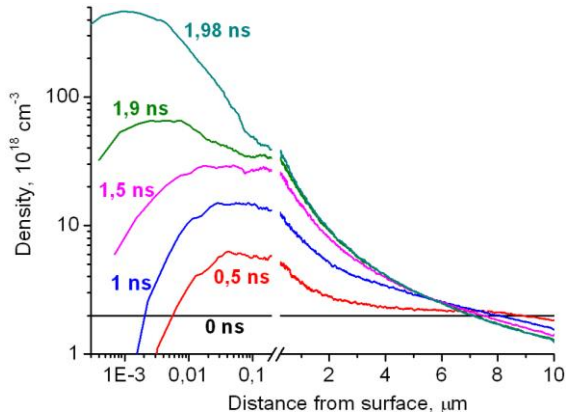


Fig. 6. Electron density along plasma domain for different time instants (# 3, Table I).

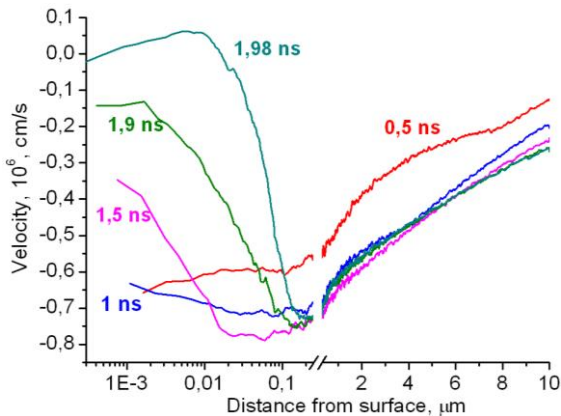


Fig. 7. Ion average velocity along plasma domain for different time instants (# 3, Table I).

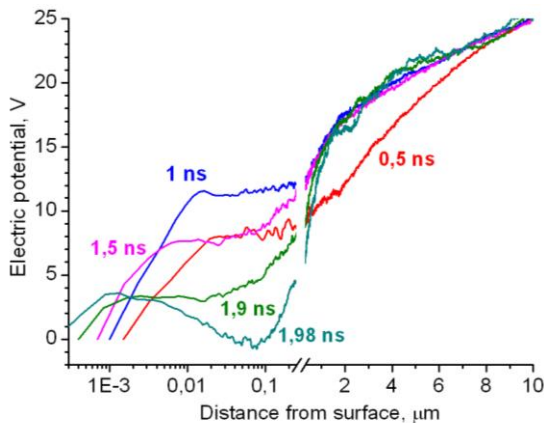


Fig. 8. Electric potential along plasma domain for different time instants. (condition 3, Table I)

plasma parameters. Due to this we have considerable flux of plasma electrons to cathode (Fig. 5). Most of the time, the voltage drop grows following the plasma density increase but remains below the floating potential. In course of temporal evolution the metal surface becomes hot enough to provide sufficiently high current density. Then, due to additional voltage fall due to the plasma resistivity (ohmic drop), the cathode drop starts to decrease again. The decrease of the cathode drop increases the plasma electron energy flux to the cathode. This in turn increases the cathode temperature. The temperature growth in turn increases the current and so on. This self-developing process leads to the temperature run-away (Fig. 2).

4. Conclusion

The computation kinetic model of the explosive emission center ignition is developed. The model treats the plasma build up and interaction between the plasma and the microp protrusion surface selfconsistently. The calculations reveal new mechanism of thermal runaway developing. It is assumed that this thermal runaway leads to the ignition of the explosive emission center. The heating of the metal surface by the flux of backward plasma electrons and the lowering of the cathode voltage drop due to current density growth play the main roles in this process.

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